Attention-Sink Signatures Dissociate During Vision–Language Pretraining

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Abstract

Attention sinks are a widely studied target of interventions in language models and are often linked, though not causally, to hallucinations in vision-language models. A sink is defined by attention concentration ($\mathrm{Sink}_1^{\epsilon}$), but in text LMs it reliably arrives with two companions: a drained value norm (v-ratio) and massive activations (h-ratio) at the same position. We track all three separately during multimodal pretraining, without assuming they are one phenomenon. The model is a 222M vision-language model with a randomly initialized decoder, position 0 being an image token, no BOS, trained under four levers: softmax attention, output-gated softmax, unnormalized sigmoid attention, and initialization from a pretrained text LM. The three signatures dissociate: each training lever produces a different combination of them, consistently across 2-3 seeds per arm. On a 1B fresh-token run (2.39 effective visual epochs, single seed), the massive-activation proxy more than doubles while attention concentration stays at exactly zero at every threshold we test. Value-norm drain emerges as a third axis, beyond the massive-activation vs. sink dissociations known from text-only models.

1 Introduction

Decoder-only transformers often allocate disproportionate attention to the first tokens of a sequence, even when their content is uninformative. These attention sinks support streaming inference [1] and have become a target of interpretation and mitigation [3]. In text language models, attention concentration often coincides with two other signatures: reduced value-vector norms [4, 6] and massive activations in the residual stream [2, 5]. Their co-occurrence motivates a basic measurement question: do all three remain coupled when a model learns from both images and text?

Existing results give reasons to distinguish the signatures. Text-model studies report joint emergence and causal links between massive activations and attention sinks [7, 8], but changes to normalization or value-path gradients can preserve attention sinks while suppressing massive activations [9, 10]. These pairwise dissociations leave open how value-norm drain varies alongside the other two quantities during multimodal pretraining.

The distinction matters when evaluating sink interventions. Reducing attention to a token does not establish that its value norm or residual activation has changed. A study measuring only concentration can therefore miss a persistent norm signature. In vision-language models, sink interventions have also been evaluated for hallucination and visual grounding [12, 13, 14, 15]. Interpreting such interventions requires identifying which internal quantity changes, then testing its relationship to behavior. Here we address the measurement step by tracking all three quantities throughout training.

Multimodal pretraining lets us study this question when the first token carries visual content. Our sequences begin with 49 image tokens and contain no BOS token, so position 0 is the first image token. Earlier multimodal studies distinguish sinks originating in vision encoders and language decoders [11, 12]. Their training-time evidence uses a pretrained decoder and follows a single activation magnitude [11]. We instead follow all three signatures from initialization in a model whose decoder starts from random weights, with a text-pretrained decoder as an inheritance control.

Preprint.

We use a 222M-parameter nanoVLM [17] to compare softmax attention, output-gated softmax, unnormalized sigmoid attention, and pretrained text initialization. The encoder is pretrained and trainable in every condition. Probes every 100 steps give a joint view of the signatures at 2-3 seeds per condition. A separate softmax run on a larger image pool extends the study to one billion tokens at 2.39 effective visual epochs, testing whether residual-norm growth without attention concentration persists under lower data repetition.

The experiments establish three results. First, the training conditions produce distinct corners of signature space, with value norms reduced or amplified depending on the condition (Figure 2, Table 1). Second, in the single-seed 1B-token run, the residual-norm ratio grows from 1.43 to 3.22 while attention concentration remains absent at every recorded probe (Section 4.2). Third, the concentration–value-norm correlation across layer/KV-group pairs changes sign between conditions, from +0.76 under baseline softmax to -0.79 under text initialization at seed 0 (Section 4.3). The contribution is joint measurement of these three signatures during multimodal pretraining, including value-norm drain beyond the two-signature dissociations established in text models [9, 10].

2 Related work

Coupled signatures in text models Gu et al. [6] connect attention concentration to small value norms and large residual activations, interpreting the sink as a key bias. Guo et al. [4] study the associated value-drain mechanism through active-dormant attention heads. Queipo-de-Llano et al. [7] link massive activations to compression valleys, layers where token representations become less diverse. Their targeted activation ablations also suppress sink formation in the studied text models. Qiu et al. [8] explain attention and residual sinks through outlier-driven rescaling, while Peng et al. [24] trace a position-zero sink circuit during text pretraining. These accounts motivate measuring the signatures together without assuming their relationship transfers unchanged to VLMs.

Dissociations and interventions Sun et al. [9] preserve attention sinks while suppressing massive activations by changing normalization. Chen, Lin and Yao [10] obtain a related dissociation with V-scale, which modifies gradients on the value path. Our study adds joint tracking of value-norm drain during multimodal pretraining. Fesser et al. [23] further distinguish a negligible-value sink that suppresses a head's update from a broadcast sink that redistributes information in trained vision transformers. Their analysis supports separating attention allocation from the value vectors it weights. For interventions, we adapt the unnormalized sigmoid attention studied by Gu et al. [6] and the attention-output gate of Qiu et al. [20]. Our gate's initialization and associated scale change are specified in Section 3.1.

Multimodal sinks and grounding Luo et al. [11] distinguish vision-encoder-propagated and decoder-emerged sinks. Their alignment-checkpoint analysis provides training-time evidence for activation growth, using a frozen vision encoder and a pretrained language decoder. Choi et al. [12] distinguish vision and language sinks and train layer-wise sink gates while keeping the underlying VLM frozen. We study a different regime, jointly tracking attention, value norms and residual norms during pretraining with random decoder initialization. High-norm tokens can also arise inside pretrained vision transformers [25], which matters because our encoder is pretrained. Work on sink-aware visual attention and decoding connects these internal patterns to grounding and hallucination [13, 14, 15]. Sink statistics have also been studied as hallucination signals in text LMs [16]. Our experiments concern the training dynamics of the signatures rather than their downstream effects.

3 Setup

3.1 Model and training conditions

All runs use a 222M-parameter nanoVLM [17]. A pretrained, trainable SigLIP-B/16 encoder [18] feeds a SmolLM2-135M-architecture decoder [19] through a learned projector (Figure 1). The decoder has $L = 30$ layers, $H = 9$ query heads per layer and $G = 3$ KV groups, giving 270 layer/query-head pairs and 90 distinct value projections. Each sequence contains 49 image tokens followed by 79 left-padded text positions. There is no BOS token. Position 0 is the first image token.

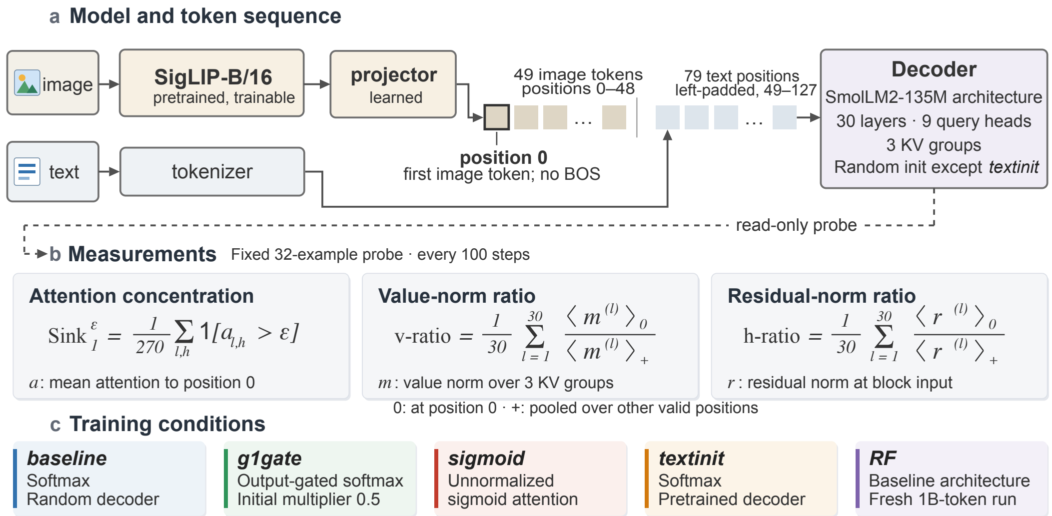

Figure 1: **Model, measurements and training conditions.** (a) A pretrained SigLIP encoder and a learned projector produce a 49-token visual prefix, followed by 79 left-padded text positions. Position 0 is the first image token and there is no BOS token. (b) A read-only probe on a fixed 32-example batch records the three signatures every 100 optimizer steps (Section 3.3). (c) The four training conditions and the fresh-data run RF. Decoder initialization is random except in textinit. The gate also halves the initial attention output and RF also changes the data stream, so neither isolates a single factor.

The four conditions share the architecture and training recipe except for the specified attention operation or decoder initialization. *baseline* uses softmax attention. *g1gate* adds the head-specific elementwise sigmoid gate of Qiu et al. [20] after attention aggregation and before the output projection. Our gate weights start at zero, making its initial output multiplier $\sigma(0) = 0.5$. *sigmoid* replaces softmax with unnormalized sigmoid attention [6]. *textinit* retains softmax and loads pretrained SmolLM2-135M decoder weights as an inheritance control. All other decoders start from random weights. The gated comparison therefore includes both learned gating and initial output scaling.

3.2 Data and training

The four-condition comparison uses the VQAv2, COCO-QA, A-OKVQA and VSR subsets of The Cauldron [21], containing 146,731 images. Table 1 compares checkpoints near 100M tokens, or 60M for textinit. Full trajectories retain the available later checkpoints. There are two seeds each for baseline and sigmoid, and three for g1gate and textinit.

At 100M tokens, training has sampled this image pool about nine times. We also train the baseline architecture on a larger FineVision stream [22], about 4.6M images, to 1B tokens at 2.39 effective visual epochs. This random-fresh run, RF, has one seed. It tests persistence under lower repetition, with dataset shift considered in Section 5. Token budgets count image tokens plus non-padding text tokens.

Training uses AdamW, weight decay 0.1, gradient clipping at 1.0, and a cosine learning-rate schedule with 3% warmup. Appendix A gives learning rates, precision, batches and validation splits. Appendix B records RF's optimizer restart and data-ordering changes.

3.3 Signature definitions

Every 100 optimizer steps, we evaluate the same 32-example probe batch and validate the probe against the model's forward pass (Appendix A). Let $a_{l,h}$ be attention to position 0, averaged over all non-padding query positions except position 0 itself, pooled across the batch. Excluding query 0 drops the trivial self-attention term that [6] includes, so our estimate is slightly stricter. Sigmoid weights are row-normalized for this diagnostic, following [6]. The model's forward pass remains unnormalized. We report raw sigmoid weights separately in Appendix F.

Concentration is the fraction of query heads exceeding a threshold,

$$\mathrm{Sink}_1^{\epsilon} = \frac{1}{LH} \sum_{l=1}^{L} \sum_{h=1}^{H} \mathbf{1}[a_{l,h} > \epsilon].$$

We use $\epsilon = 0.3$ [6] and check 0.2 and 0.4. Table 1 uses 0.2, a lower threshold that makes the absence of concentration harder to establish. Figure 2 instead uses $a_{\max} = \max_{l,h} a_{l,h}$, the continuous maximum attention across all 270 pairs.

Norm ratios compare position 0 with the remaining valid image and text positions. For sample b, position i, layer l and KV group g, define $m_{b,i}^{(l)} = G^{-1} \sum_{g=1}^{G} \|v_{b,i}^{(l,g)}\|_2$ and $r_{b,i}^{(l)} = \|h_{b,i}^{(l)}\|_2$. Values are measured after the normalized input's value projection. Residuals are measured at the block input. Let $\langle\cdot\rangle_0$ denote the batch mean at position 0 and $\langle\cdot\rangle_+$ the pooled mean over all other non-padding positions. Then

$$\text{v-ratio} = \frac{1}{L} \sum_{l=1}^{L} \frac{\langle m^{(l)} \rangle_0}{\langle m^{(l)} \rangle_+}, \qquad \text{h-ratio} = \frac{1}{L} \sum_{l=1}^{L} \frac{\langle r^{(l)} \rangle_0}{\langle r^{(l)} \rangle_+}.$$

A v-ratio below 1 indicates relative value-norm drain, and above 1 amplification. The main v-ratio averages 30 layer ratios after pooling the three KV-group norms within each layer. Section 4.3 separately retains all 90 group ratios. The h-ratio measures residual-norm asymmetry and serves as a proxy for channel-level massive activations [2, 5]. Appendix E checks how position-0 measurements relate to extrema elsewhere in the sequence.

4 Results

4.1 Each training condition lands in a different corner of signature space

Table 1 summarizes per-seed ranges at checkpoints near 100M tokens, or 60M for textinit (Appendix D). The clearest separation is between sigmoid and textinit: both show strong relative concentration, but sigmoid amplifies value norms and retains an h-ratio near 1, whereas text initialization combines value drain with a much larger h-ratio. Figure 2 shows the corresponding trajectories.

Table 1: **The four corners, ranges over 2-3 seeds per condition.** Concentration uses $\mathrm{Sink}_1^{0.2}$. The v-ratio and h-ratio are the layer-averaged quantities defined in Section 3.3.

arm	concentration	value-norm ratio	residual-norm ratio
baseline	absent (0.000)	drain (0.69–0.72)	1.7–2.2
g1gate	near-absent (0.004–0.011)	drain (0.81–0.85)	1.7–2.2
sigmoid	strong (0.76–0.83)	amplified (1.48–1.60)	1.1–1.3
textinit	strong (0.56–0.85)	drain (0.38–0.63)	5.5–42.5

The g1gate and baseline corners overlap on residual-norm ratio and both have little concentration. Their value-norm ranges separate, with less drain in the gated condition. This difference belongs to the combined gate-and-initial-scale intervention in Section 3.1. It does not isolate a benefit of learned gating. Under sigmoid, high normalized concentration coexists with decreasing raw attention weights to position 0 (Appendix F). Thus, the relative attention and norm signatures separate even when the absolute attention scale changes.

The textinit corner repeats across seeds, while its h-ratio varies from 5.5 to 42.5, with median 12.2. Per-position scans also show that attention and norm extrema can occupy different tokens (Appendix E). Figure 3 shows a first-position attention preference already present before multimodal training in textinit, consistent with inheritance from the text decoder. This preference is subthreshold at $\epsilon = 0.3$. Threshold-crossing order and spatial dissociations are detailed in Appendix I. Appendix G outlines hypotheses for the corners, and Appendix H summarizes metric provenance.

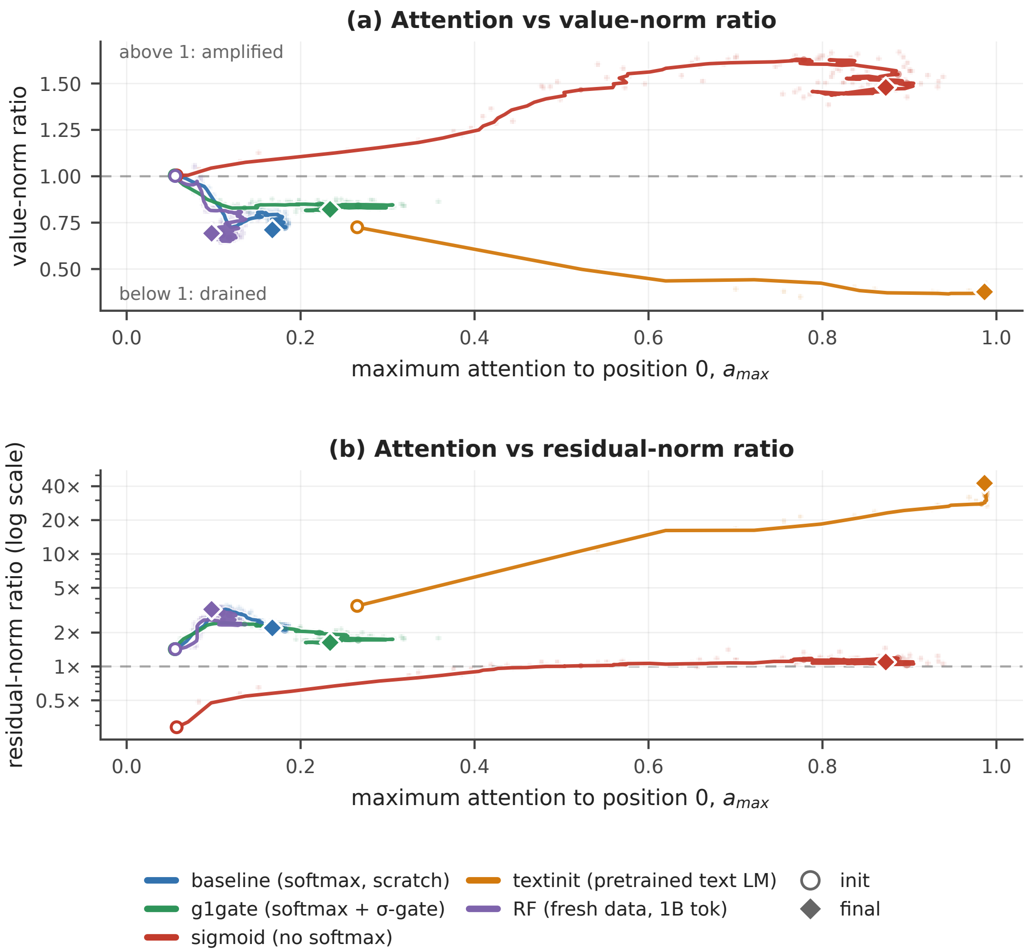

Figure 2: **Signature trajectories during pretraining.** (a) Value-norm ratio and (b) residual-norm ratio, on a logarithmic vertical scale, against maximum attention $a_{\max}$ (Section 3.3). Open circles mark initialization and diamonds the final recorded checkpoints: 174M tokens for baseline, 103M for g1gate, 102M for sigmoid, 60M for textinit, and 1B for RF. All paths use seed 0. Faint points are raw probes. Lines are smoothed (Appendix F). The dashed horizontal reference is a norm ratio of 1.

4.2 Residual-norm growth without concentration over 1B tokens

The repeated-data comparison shows a substantial validation gap, with loss near 0.44 on training images paired with new questions versus 1.18 on held-out images. RF tests whether the absence of concentration persists with less image reuse. Its held-out loss ends at 0.638 and has a negative fitted slope over the second half of training, despite evaluation-to-evaluation fluctuations (Appendix A).

Across RF's 698 recorded probes, none of the 270 heads exceeds 0.2 mean attention to position 0. The largest value observed is 0.151. Consequently $\mathrm{Sink}_1^{\epsilon}$ is exactly zero at all probes for $\epsilon \in \{0.2, 0.3, 0.4\}$. Over the same run, the residual-norm ratio rises from 1.43 to 3.22 (Table 2). The result is a growing norm asymmetry without thresholded attention concentration on the fixed probe batch.

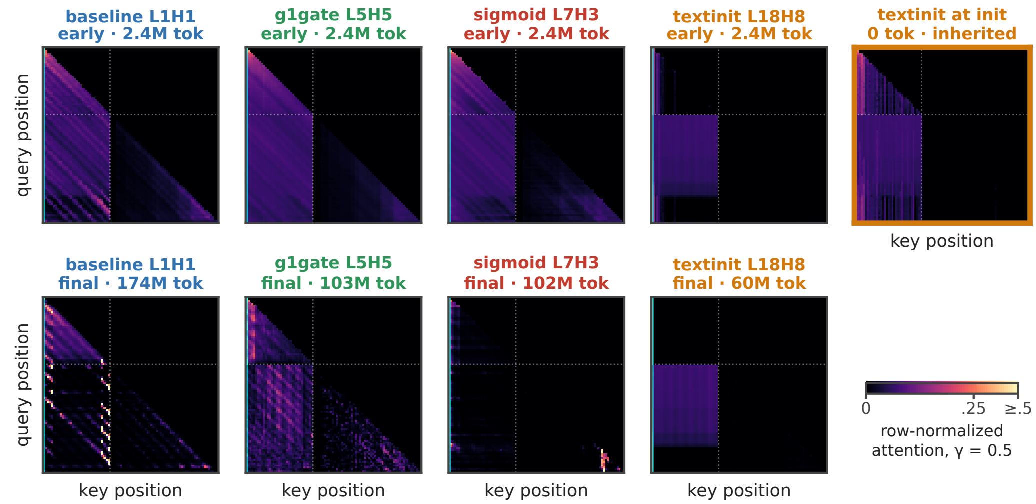

Figure 3: **First-position attention across training conditions.** Selected-head query-by-key maps at early and final checkpoints, plus textinit at initialization, all seed 0. Each map is a batch-mean attention matrix normalized by row. The color scale saturates at 0.5. Cyan marks key position 0, dotted lines the image/text boundary. Heads are selected separately by condition. These qualitative maps use the batches and selection procedure documented in Appendix F.

Table 2: **RF, one seed, initialization to 1B tokens.**

signature	initialization	1B tokens	change
$\mathrm{Sink}_1^{0.3}$	0.000	0.000	zero at all 698 probes
maximum attention $a_{\max}$	0.056	0.098	remains below 0.2
h-ratio	1.43	3.22	2.25-fold increase
v-ratio	1.00	0.69	net drain

The h-ratio rises from 2.40 near 57M tokens to 3.22 at 1B, so the increase continues well beyond the initial warmup. Figure 2b shows this change at low concentration. The value-norm ratio follows a different trajectory: about 75% of its net decline occurs by 57M tokens, followed by partial recovery. It therefore provides evidence of persistent drain, not a monotonic increase in drain throughout training.

4.3 Concentration–value-norm correlations change sign across conditions

Table 3 reports Pearson correlations between attention to position 0 and value-norm ratio over 90 layer/KV-group pairs per condition at seed 0. Each group's attention is averaged over its three query heads. Its value ratio is counted once (Figure 5, Appendix C). The correlation is positive under baseline softmax and negative under text initialization.

Table 3: **Pearson r across 90 layer/KV-group pairs at the final available reprobe, seed 0.**

baseline	g1gate	sigmoid	textinit	RF	pooled
+0.76	+0.57	−0.03	−0.79	+0.51	−0.20

These are descriptive associations within trained models. Layer dependence and the single seed per condition preclude treating the 90 observations as independent experimental replicates. The sign change shows that greater first-position attention is associated with larger values in one condition and smaller values in another. It does not establish statistical independence or rule out a nonlinear or layer-dependent relationship. The pooled value combines within-condition and between-condition variation and is not evidence that the signatures are unrelated.

5 Limitations

Measurement scope The h-ratio measures residual-norm asymmetry. Channel-level outlier statistics are needed to establish massive activations in the stricter sense of [2, 5]. All headline metrics use one fixed 32-example probe batch and position 0. Their denominators include both image and text positions. Supplementary image-only norm profiles retain the RF asymmetry at the inspected checkpoint, while spatial scans show that textinit's attention and norm extrema can separate (Appendix E). The reported textinit ratios are position-specific measurements, not estimates of each run's largest possible asymmetry.

Training controls The four conditions are not a factorial experiment. In g1gate, learned gating and the initial 0.5 output scale are inseparable without a scale-matched control. The vision encoder is pretrained and trainable in every run. Increasing residual norms may involve adaptation of inherited visual structure as well as decoder learning. A random-encoder control would test that contribution. Textinit also differs in prior training and token budget, so its lower validation loss is not evidence for a superior attention intervention.

RF and generalization RF provides one seed at 1B tokens. It includes a weights-only restart near 57M tokens and a smaller shuffle buffer later in training (Appendix B). Changing from The Cauldron to FineVision reduces repetition but also changes the training distribution. Its probe remains on The Cauldron, and no separate seen-image validation split is available. The study uses one 222M architecture and 128-position sequences. Later sink emergence, larger models and other sequence layouts remain untested. We evaluate internal signatures, without downstream grounding or hallucination benchmarks.

6 Conclusion

In text language models the three attention-sink signatures arrive together. However, in a vision-language model with a randomly initialized decoder, they need not. Tracked separately, they come apart under each training lever applied. Across four interventions the arms occupy separate corners of signature space, with value norms strongly drained, mildly drained or amplified. The sharpest case is the fresh-data run, where over one billion tokens the massive-activation proxy more than doubles while attention concentration stays at exactly zero. This extends the two-way dissociations of text-only models [9, 10] to a separately measured third axis, value-norm drain, in multimodal pretraining. Massive activations can causally produce sinks in text LMs [7]. Here the coupling can also fail to form.

Practical implications No single signature acted as a reliable proxy for the others in our arms. A model without a concentration sink can still develop a growing residual-norm asymmetry, and an intervention judged successful by one metric may leave the other signatures unchanged. A sink mitigation for VLMs should therefore report all three. Their downstream relevance, for grounding or hallucination, requires behavioral evaluation.

Status and next steps This is work in progress at small scale, and Section 5 names the gaps. Next steps are a randomly initialized vision encoder to isolate the decoder's contribution to the residual-norm signal, a second fresh-data seed and fresh-data runs past 1B tokens, a scale-matched gate control to separate gating from the half-scale initialization confound, and channel-level statistics to turn the h-ratio proxy into a measurement of massive activations.

Reproducibility Appendix A specifies the self-validating probe and the training recipe, and Appendix D lists per-seed results. We will release the code, run configurations, logs and checkpoints upon acceptance. Identifying links are withheld for double-blind review.

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A Extended methodology

Probe. The function `probe_sinks()` recomputes the decoder in eager mode, fp32, without gradients. Every call compares its final hidden states with the model's forward pass: the maximum absolute difference on valid positions divided by the maximum absolute reference value must be below 0.01. Training uses fused attention, bf16 autocast and `torch.compile`. The diagnostic excludes padding queries and query position 0. Attention averages pool valid queries across examples. Norms are averaged before ratios, as specified in Section 3.3.

Optimization. AdamW uses weight decay 0.1 and gradient clipping at 1.0. Learning rates are 4×10^{-4} for the decoder, 2×10^{-3} for the projector and 10^{-4} for the encoder, with cosine decay and 3% warmup. Batch size is 128. The gated condition adds a zero-initialized output gate, and textinit imports pretrained decoder weights.

Token accounting. Budgets count 49 image tokens plus the non-padding text tokens in each example. A step of 128 examples therefore contributes at most 16,384 tokens and about 9.5K in the repeated-data runs. These are processed tokens, not a count of unique examples or solely the tokens used in the text loss.

Probe batches. Live training trajectories use a fixed 32-example batch from the repeated The Cauldron tail, including for RF. Saved-checkpoint reprobes provide additional per-group and per-position arrays. The local streaming reprobe constructs a different 32-example batch with seed 0. Figure 3 uses batch-mean saved matrices, with its sigmoid column regenerated on that streaming batch (Appendix F). Results from different probe products are identified separately rather than treated as numerically identical.

Validation. The held-out image pool contains 1,024 examples. Every 500 steps, loss is averaged over its first 512 examples in four fixed batches of 128. Repeated-data runs also evaluate training images with fresh question–answer text (`val_seen`). RF's seen-image loader reuses its held-out pool, so only `val_unseen` is reported. RF ends at 0.638. Its mean held-out loss falls from 1.35 over the first ten evaluations to 0.68 over the last ten, with a negative fitted slope in the second half.

LLM assistance. Large-language-model coding assistants were used to write and refactor the training, probe and analysis code, and to edit prose. The experimental design, the measurements, the analysis decisions and the claims are the authors' own. The released repository keeps its commit history, including the assistant-authored commits.

B Extended limitations

B.1 Position and norm aggregation

Headline measurements anchor on the first image token. Appendix E checks attention mass across positions and reports a supplementary norm scan. Textinit's attention and norm extrema need not coincide, and their locations vary by seed. The position-0 ratios therefore characterize that position. Relocating the measurement to the attention maximum does not necessarily increase the residual-norm ratio. Main norm ratios compare image position 0 with a mixed image/text denominator, while the supplementary image-only profiles use a different aggregation described in Appendix E.

B.2 Gating and initialization

Our gate begins at $\sigma(0) = 0.5$, halving the attention output before the output projection. The measured difference from baseline combines this initial scale change with subsequent learned gating. A constant half-scale control would distinguish the two explanations. The present comparison cannot do so.

B.3 Pretrained encoder

The SigLIP encoder is pretrained and trainable throughout. High-norm visual tokens [25] or propagated visual sinks [11] may contribute to decoder-side signatures. RF's rising h-ratio establishes change during multimodal training, but cannot distinguish new decoder structure from amplification or adaptation of inherited representations. Random-encoder and frozen-encoder controls would separate these possibilities.

B.4 Seed variation

Textinit's h-ratio spans 5.5–42.5 at the reported checkpoints. Strong concentration and value drain recur across seeds, but their magnitudes and spatial locations vary. The study reports ranges and individual measurements rather than treating seed 0 as representative of the arm's magnitude.

B.5 RF restart and provenance

RF contains one weights-only restart near 57M tokens after an out-of-memory failure: weights were restored and AdamW moments discarded. Recorded v-ratio and h-ratio values agree at the shared checkpoint, and concentration is zero on both sides. The restart remains part of the training trajectory. Main per-seed tables combine archived seed-0 summaries with independently consolidated later seeds. Figures 2 and 4 retain the available seed-0 probe trajectories. Raw checkpoint reprobes support the per-group analysis separately from those archived table summaries.

B.6 Data and evaluation scope

RF reduces repetition by changing the training mixture to FineVision. The estimated overlap with the repeated subsets is under 3% by configured dataset composition, without image-level deduplication. Its shuffle buffer decreases from 1,500 to 500 during training for memory reasons. Domain shift, this ordering change and the optimizer restart prevent interpreting RF as an isolated causal test of repetition. RF's signatures are evaluated on The Cauldron probes, so the absence result is specific to that evaluation distribution. A matched fresh/repeated FineVision comparison and a second RF seed are direct follow-ups.

C Supporting figures

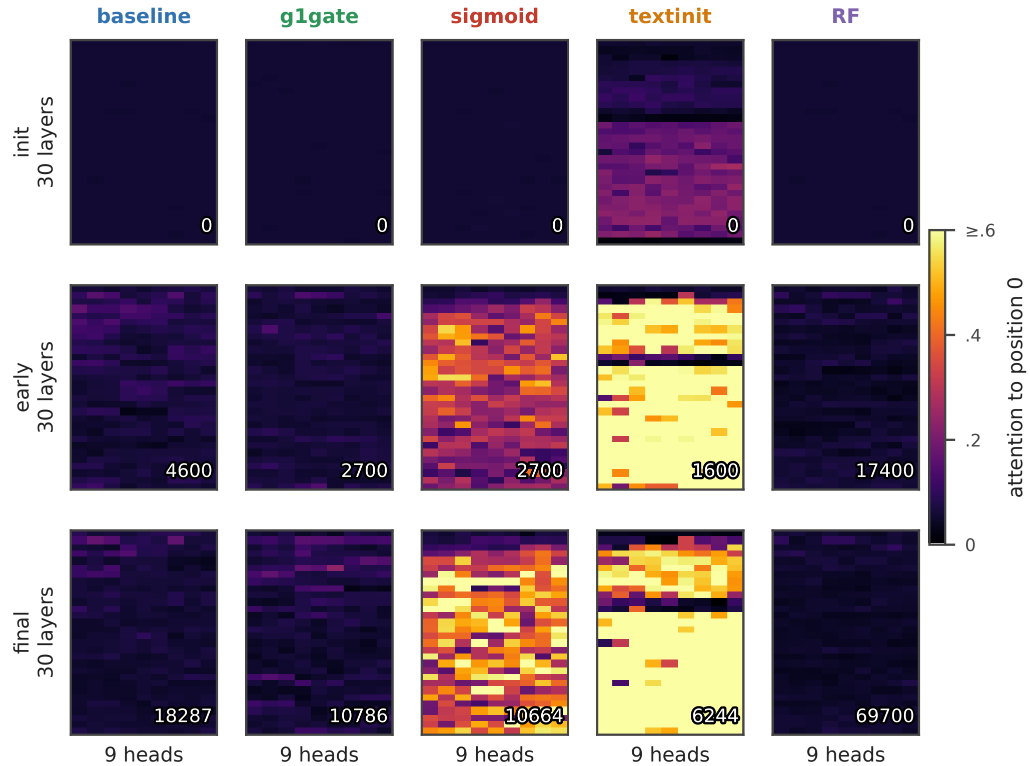
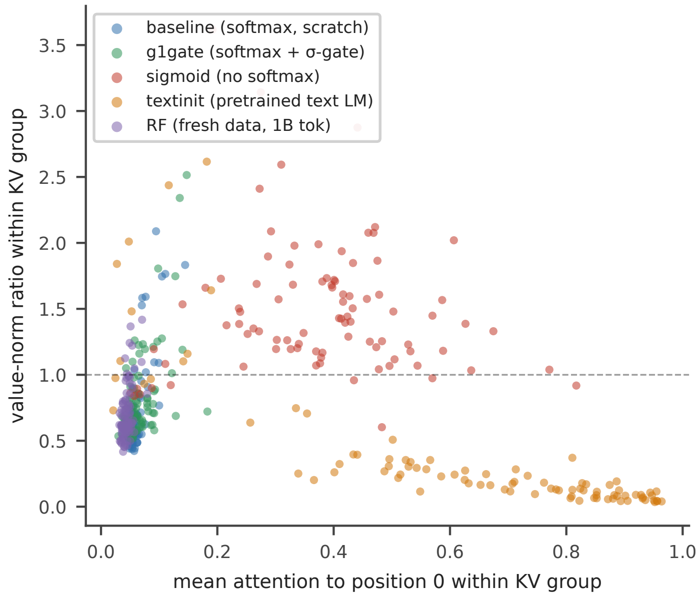


Figure 4: **First-position attention by layer and query head.** Initialization, an intermediate checkpoint near one quarter of each run's steps, and the final probe, seed 0. Rows within each panel are layers and columns query heads. Inset numbers are optimizer steps. Attention is row-normalized for sigmoid. All panels share a scale saturated at 0.6. Run endpoints are those of Figure 2.



arm	n	Pearson r
baseline	90	+0.76
g1gate	90	+0.57
sigmoid	90	-0.03
textinit	90	-0.79
rf	90	+0.51
POOLED	450	-0.20

90 KV groups
per condition
seed 0

Descriptive r

Figure 5: **Concentration–value-norm associations by condition.** Each point is one layer/KV-group pair, 90 per condition, at the final available seed-0 reprobe. Attention averages the group's three query heads. Its value ratio is counted once. Pearson correlations match Table 3. The observations are descriptive and retain dependence between groups within a layer.

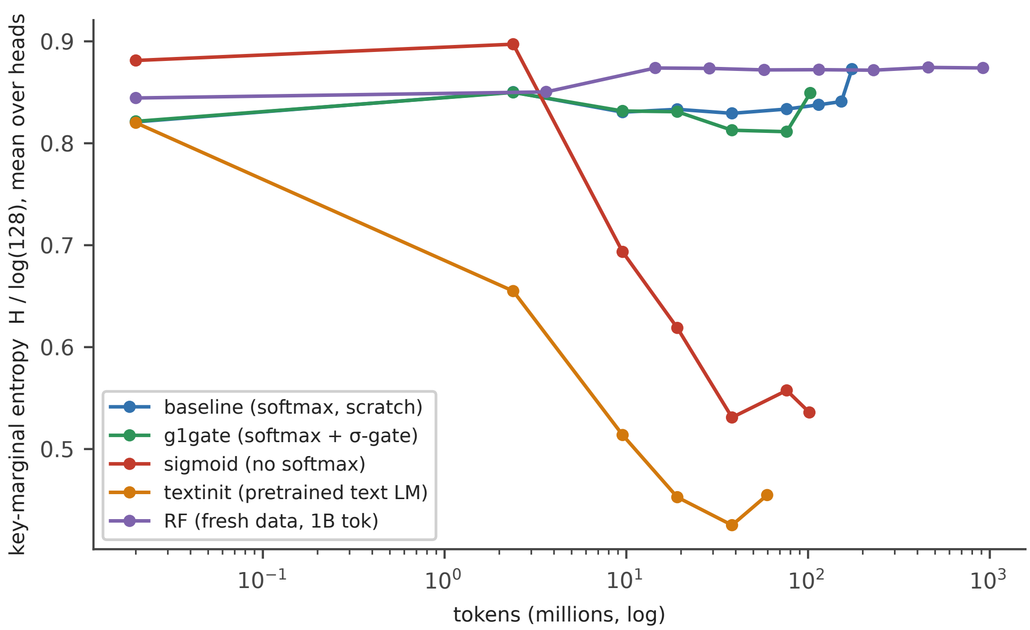


Figure 6: **Entropy of attention over key positions.** Shannon entropy of the query-averaged key distribution, normalized by log(128), then averaged over heads and layers, seed 0. This is marginal entropy, not the average entropy of individual attention rows. Checkpoints are joined by lines. Initialization is placed at 0.02M tokens for the logarithmic axis.

D Full per-seed signature table

Table 4 provides the values summarized in Table 1. Seed-0 entries use the archived matched-budget table, while Figure 2 uses full live trajectories, including baseline's later 174M endpoint. A dash denotes a maximum not retained in this table's source. It does not mean the metric is unavailable in the trajectory logs.

Table 4: **Per-seed signatures at the comparison checkpoints.**

arm	seed	tokens	$\text{Sink}_1^{0.2}$	mean attn→pos0	max attn→pos0	v-ratio	h-ratio
baseline	s0	101M	0.000	0.068	—	0.723	2.16
baseline	s1	100M	0.000	0.062	—	0.687	1.71
g1gate	s0	101M	0.004	0.068	—	0.805	1.67
g1gate	s1	100M	0.011	0.073	~0.21	0.845	2.04
g1gate	s2	100M	0.0037	0.072	~0.22	0.854	2.24
sigmoid	s0	102M	0.830	0.377	—	1.479	1.10
sigmoid	s1	100M	0.756	0.311	—	1.603	1.30
textinit	s0	60M	0.852	0.627	—	0.377	42.5
textinit	s1	60M	0.556	0.235	~0.53	0.634	5.50
textinit	s2	60M	0.578	0.232	~0.59	0.484	12.2

The textinit h-ratio is much larger at seed 0 than at seeds 1 and 2. This variation persists in the later supplementary norm profiles, which also show spatial differences (Appendix E.1). Small g1gate concentration fractions indicate a few heads above 0.2. Some seed-0 probes also cross 0.3, as shown in Figures 7 and 8.

E Per-position attention and norm profiles

Table 5 reports raw attention weights from the final available seed-0 checkpoint reprobes, averaged over valid queries and the batch. Softmax weights are normalized. Sigmoid weights are raw and unnormalized. The sigmoid row therefore describes absolute gate weights, distinct from the normalized concentration in Table 1.

Table 5: **Raw first-position attention in checkpoint reprobes, seed 0.**

arm	mean at pos0	maximum head at pos0	position with greatest mean weight
baseline	0.059	0.165	0
g1gate	0.068	0.234	0
sigmoid	0.301	0.659	0
textinit	0.627	0.986	0
RF	0.044	0.094	1

E.1 Spatial checks and image-only norm comparison

The supplementary norm scan records extrema within the first 20 image positions. Table 6 reports that scan alongside the attention-profile checks. Dashes denote quantities not included for those seeds. These positions are scan extrema, not proven global extrema over all 128 positions.

Table 6: **Positions of attention maxima and norm extrema in supplementary scans.**

run	seed	attention maximum	residual maximum	value minimum
baseline	s0	0	0	1
baseline	s1	0	—	—
g1gate	s0	0	6	6
g1gate	s1, s2	0	—	—
sigmoid	s0	0	14	13
sigmoid	s1	0	—	—
textinit	s0	0	0	0
textinit	s1	0	1	5
textinit	s2	1	13	13
RF	s0	1	0	0

At RF step 64,000, the image-only residual-norm ratio is 2.516 and the value-norm ratio 0.623. These supplementary ratios first average norms over layers and, for values, KV groups, then divide position 0 by the median across image positions 1–48. They retain the qualitative norm asymmetry with an image-only reference, but are not numerically interchangeable with the main layer-averaged, mixed-position ratios. The scan has no matched initialization profile and does not independently establish the twofold growth.

For textinit seed 1, the attention maximum stays at position 0 while the residual maximum and value minimum occur at positions 1 and 5. At seed 2, attention peaks at position 1 and both norm extrema at position 13. Measuring at the attention maximum in the seed-2 supplementary profile gives an h-ratio of 9.46, below the position-0 ratio of 14.70. Spatial separation therefore does not make the reported position-0 ratios lower bounds on the norm ratio at the attention sink.

F Measurement and rendering details

Figure 2 trajectories Moving-average windows contain five probes horizontally and nine vertically. Faint points show unsmoothed values. Open circles and final diamonds use raw endpoints. All conditions use seed 0 and their full available trajectories. The horizontal maximum and the thresholded fraction in Table 1 are different summaries of the same head-level attention scores.

Group correlations Table 3 uses the final available per-head reprobes: baseline step 18,287; g1gate 10,786; sigmoid 10,664; textinit 6,244; RF 64,000. The RF reprobe precedes its final live probe. Value ratios are retained per KV group, with attention averaged over the group's query heads. Uncollapsed query-head correlations are +0.67, +0.53, −0.03, −0.76 and +0.43 respectively. Collapsing preserves their signs. Neither aggregation removes layer dependence.

Raw and normalized sigmoid weights Gu et al. [6] also row-normalize proxy weights when measuring sinks in models trained with unnormalized sigmoid attention. Our normalized diagnostic is consistent with that distinction between measurement and forward computation. In the saved sigmoid reprobe, head L7H3 has mean normalized first-position attention 0.873 at step 10,664, raw first-position weight 0.065, and mean total raw row weight 0.110. At step 250, the corresponding raw quantities are 0.309 and 6.44. The growing relative concentration thus accompanies a shrinking absolute weight budget. The ratio of mean raw weights, about 0.59 at the endpoint, differs from the mean normalized attention, 0.873, because averaging and division do not commute.

Figure 3 batches and selection The panels show batch-mean matrices over 128 positions: 49 image positions followed by left-padded text. They do not correspond to a single prompt. Each matrix is normalized by row after batch averaging, whereas the headline probe averages normalized weights over valid queries. Scalar headline measurements are therefore not overlaid on the qualitative matrices.

For baseline, g1gate and textinit, we choose the higher final-checkpoint first-position share among the two heads retained in the original matrix dump, then hold that head fixed across displayed checkpoints. These are L1H1, L5H5 and L18H8 respectively, using zero-based indices. The sigmoid panels use L7H3, selected using normalized attention in the checkpoint reprobe and regenerated from a streaming batch. The exact decoded text strings were not retained with the matrix archive.

Bright off-position-0 cells show query-specific attention to other keys. For example, the final sigmoid map has strong cells in the text-to-text region. Such isolated cells do not by themselves establish a persistent sink or identify the semantic role of the attended tokens. Padding and batch averaging also affect the qualitative display. The common power-law color scale uses exponent 0.5 and saturates at 0.5.

G Mechanistic hypotheses

The following explanations motivate future controls. They are not established by the signature measurements.

An output gate could reduce the need to suppress a value vector, consistent with the milder value drain in g1gate. A scale-matched ungated control is needed to distinguish that explanation from initial output scaling. Under sigmoid attention, removing the sum-to-one constraint permits a small total attention update. Testing how this relates to value amplification requires measuring effective updates as well as norms. Text initialization may transfer a first-position circuit to the visual prefix. Ablating that circuit before alignment would test its role in the inherited profile.

H Metric provenance

The main per-seed table, live trajectories and checkpoint reprobes are distinct products. Mean attention, maximum attention and the fraction above a threshold are reported under separate names. Positional attention profiles use weights over the full sequence, without renormalizing a displayed slice. Position-0 norm ratios remain labeled by their anchor even when supplementary scans find larger norms elsewhere.

I Threshold ordering and spatial dissociation

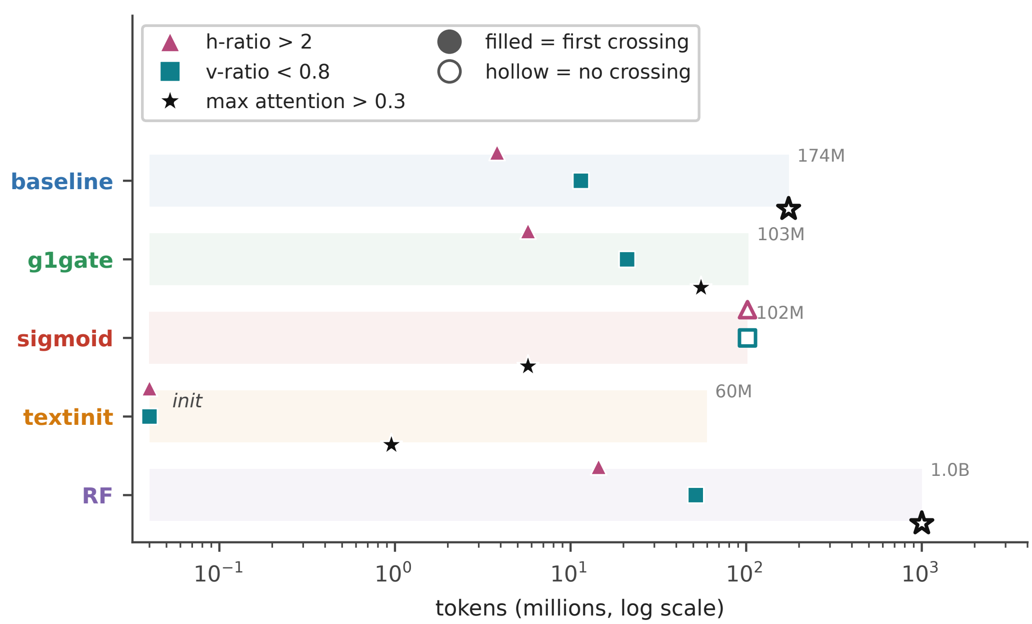


Figure 7: **First threshold crossings during training.** Filled markers indicate the first recorded crossing of h-ratio > 2, v-ratio < 0.8, or maximum attention > 0.3. Hollow markers mark conditions with no observed crossing by the end of the track. Initialization is placed at 0.04M tokens on the log axis. Crossing times are interval-censored by the 100-step probe cadence. All runs use seed 0.

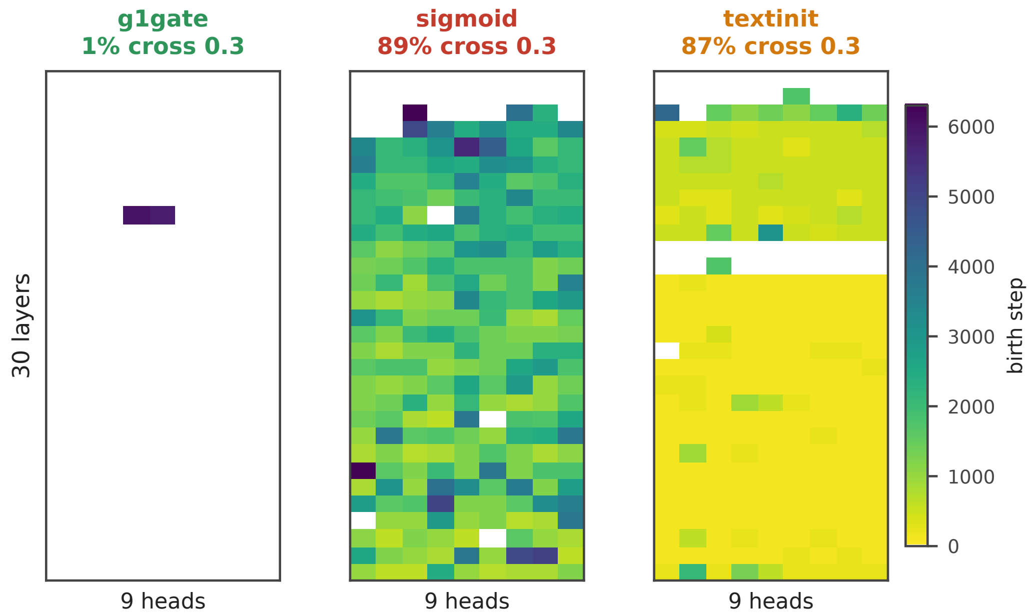


Figure 8: **First concentration crossing by layer and query head.** Colors show the first step with mean attention to position 0 above 0.3. White cells never cross during the recorded trajectory. The fractions crossing at least once are 1% for g1gate, 89% for sigmoid and 87% for textinit. Baseline and RF have no crossing heads. Layers increase downward and query-head indices increase to the right.

Ordering In baseline, residual-norm ratio first exceeds 2 at 3.82M tokens and value-norm ratio falls below 0.8 at 11.45M, with no concentration crossing. G1gate crosses the norm thresholds at 5.72M and 20.98M, then crosses concentration at 55.32M in a small number of heads. Sigmoid crosses concentration at 5.72M without crossing either norm threshold. Textinit starts above the residual threshold and below the value threshold, then crosses concentration by 0.95M tokens. RF crosses the norm thresholds at 14.44M and 51.64M without crossing concentration. These are first observed events, not guarantees of sustained crossings.

The norm extrema also separate spatially in textinit (Appendix E.1). Together with the temporal patterns, this supports tracking each signature with its own threshold, position and aggregation, rather than inferring one from another.

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